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# Extremal problems on the $p$-Seidel energy of graphs

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**Abstract**

Let $G$ be a graph with vertex set $\{v_1, \ldots, v_n\}$. The *Seidel matrix* $S(G)$ is the $n \times n$ symmetric matrix with zero diagonal whose $(i, j)$-entry, for $i \neq j$, is $-1$ if $v_i$ and $v_j$ are adjacent and 1 otherwise. If $\lambda_1, \ldots, \lambda_n$ are the Seidel eigenvalues of $G$, then the $p$-Seidel energy of $G$ is defined by

$$\mathcal{E}_p(S(G)) = \sum_{i=1}^{n} |\lambda_i|^p, \qquad p > 0.$$

For $p > 2$, we establish a universal lower bound for the $p$-Seidel energy of graphs of order $n$ and characterize the equality case in terms of symmetric conference Seidel matrices. Consequently, whenever a symmetric conference matrix of order $n$ exists, this bound determines the minimum $p$-Seidel energy and all graphs attaining it. For $0 < p < 2$, we obtain the corresponding upper bound and equality characterization; hence, whenever a symmetric conference matrix of order $n$ exists, the maximum $p$-Seidel energy is determined. In addition, for $p > 2$, $n \equiv 1 \pmod 4$, and $r = \frac{n-1}{2}$, we establish a lower bound for the $p$-Seidel energy among all $r$-regular graphs of order $n$ and characterize its equality case. Consequently, whenever a conference graph of order $n$ exists, this bound determines the minimum $p$-Seidel energy and all graphs attaining it. In particular, when $n$ is a prime power with $n \equiv 1 \pmod 4$, the Paley graph $P(n)$ is a conference graph and hence attains this minimum. For $p \geq 3$ and $n = 2r$, we determine the maximum $p$-Seidel energy among all $r$-regular graphs of order $n$ and characterize the equality case. We also determine the maximum $p$-Seidel energy among $r$-regular graphs with $n = 2r+2$ for $p \geq 3$ and the maximum is attained uniquely by the disjoint union of two complete graphs of order $r + 1$. Finally, we pose several open problems concerning the $p$-Seidel energy for different values of $p$.

**Keywords.** Spectral graph theory; Seidel matrix; $p$-Seidel energy; Complete graph; Conference graph; Paley graph.
**Mathematics Subject Classifications:** 05C50

## 1 Introduction

Let $G = (V, E)$ be a simple graph with $n$ vertices ($n$ is called the order of $G$) and $m$ edges ($m$ is called the size of $G$) having vertex set $V = \{v_1, v_2, \ldots, v_n\}$ and edge set $E = \{e_1, e_2, \ldots, e_m\}$. The adjacency matrix $A(G) = [A_{ij}]_{n\times n}$ of $G$ is a $(0, 1)$-square matrix of order $n$ whose $ij$-entry is equal to 1 if $v_i$ is adjacent to $v_j$ and equal to 0 otherwise. The adjacency matrix of a graph is a real symmetric matrix. The *degree* of a vertex $i \in V$ in

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a graph $G$ is the number of vertices adjacent to $i$. The *complement* of a graph $G$ is the graph on the same set of vertices in which two distinct vertices are adjacent if and only if they are not adjacent in $G$, and is denoted by $\overline{G}$. The *complete graph* of order $n$ is the graph with $n$ vertices in which every pair of distinct vertices is adjacent, and is denoted by $K_n$. A graph $G$ is said to be *$r$-regular* if each vertex of $G$ has degree $r$. A graph $G$ is *complete bipartite* if $V = X \sqcup Y$ with $X, Y \neq \varnothing$ and $v_i v_j \in E$ for all $v_i \in X$, $v_j \in Y$, and no edges join two vertices within $X$ or within $Y$. When $|X| = |Y|$, the graph is called *balanced*. If $|X| = a$ and $|Y| = b$, the graph is denoted by $K_{a,b}$. Throughout this article, all graphs are non-empty, undirected, simple, finite, and have at least one edge. Whenever connectedness is required, it will be stated explicitly.

Let $M$ be a Hermitian matrix of order $n$. Then $M$ has $n$ real eigenvalues, denoted by $\alpha_1(M), \alpha_2(M), \ldots, \alpha_n(M)$. The *$p$-energy* of $M$, denoted by $\mathcal{E}_p(M)$, is defined to be the sum of the $p$th powers of the absolute values of the eigenvalues of $M$ i.e.,

$$\mathcal{E}_p(M) = \sum_{i=1}^{n} |\alpha_i|^p, \quad p > 0.$$

The well-known concept of the *energy* of a graph $G$, denoted by $\mathcal{E}(G)$, is $\mathcal{E}_1(A)$, where $A = A(G)$ is the adjacency matrix of $G$. In 2025, Akbari, Kumar, Mohar, and Pragada studied the positive and negative square energies of a graph [2]. The *Seidel matrix* $S(G) = [S(G)_{ij}]_{n\times n}$ of a graph $G = (V,\ E)$ is a matrix of order $n$ defined by:

$$S(G)_{ij} = \begin{cases} 0, & v_i = v_j, \\ -1, & v_i \neq v_j \text{ and } v_i v_j \in E, \\ +1, & v_i \neq v_j \text{ and } v_i v_j \notin E. \end{cases}$$

In 1966, van Lint and Seidel introduced the concept of the Seidel matrix for graphs and studied its connection with equiangular lines in [13]. The Seidel matrix can also be written as $S(G) = J - I - 2A(G)$, where $A(G)$, $J$, and $I$ are the adjacency matrix of $G$, the matrix with all entries 1, and the identity matrix, respectively. For more details, see [1, 8, 9].

Let $\lambda_1, \lambda_2, \ldots, \lambda_n$ be the Seidel eigenvalues of $G$. The notation $\mathbf{1}$ is the vector $(1, 1, \ldots, 1)^\top \in \mathbb{R}^n$. The notations $\mathrm{tr}(M)$ and $\mathrm{Spec}(M)$ are the trace and collection of eigenvalues of the matrix $M$, respectively. Suppose that $G = (V,\ E)$ has $q$ distinct Seidel eigenvalues. Then the Seidel spectrum of $G$ can be written as

$$\mathrm{Spec}(S(G)) = \{\underbrace{\lambda_1, \ldots, \lambda_1}_{m_1 \text{ times}}, \underbrace{\lambda_2, \ldots, \lambda_2}_{m_2 \text{ times}}, \ldots, \underbrace{\lambda_q, \ldots, \lambda_q}_{m_q \text{ times}}\},$$

where $m_i$ is the algebraic multiplicity of $\lambda_i$, for $1 \leq i \leq q$ with $\sum_{i=1}^{q} m_i = n$.

In 2020, Akbari, Einollahzadeh, Karkhaneei, and Nematollahi defined the *$p$-Seidel energy* of $G$ by

$$\mathcal{E}_p(S(G)) = \sum_{i=1}^{n} |\lambda_i|^p, \quad p > 0,$$

where $|\lambda_i|$ denotes the absolute value of $\lambda_i$ [1, Page 2]. For $p = 1$, the 1-Seidel energy is the usual Seidel energy of $G$, denoted by $\mathcal{E}(S(G))$.

**Definition 1.1.** Let $G = (V, E)$ be a graph, and let $V = V_1 \sqcup V_2$ be a partition of its vertex set. The *Seidel switching* of $G$ with respect to $V_1$ (equivalently, with respect to the partition $(V_1, V_2)$) is the graph $G'$ obtained from $G$ as follows: the adjacency relation

between $V_1$ and $V_2$ is reversed, while the adjacency relations within $V_1$ and within $V_2$ remain unchanged. More precisely, for $u \in V_1$ and $v \in V_2$,

$$uv \in E(G') \iff uv \notin E(G),$$

whereas for two vertices belonging to the same part,

$$uv \in E(G') \iff uv \in E(G).$$

Let $D = \mathrm{diag}(d_1, \ldots, d_n)$, where $d_i = \begin{cases} 1, & v_i \in V_1, \\ -1, & v_i \in V_2. \end{cases}$ If $G'$ is obtained from $G$ by the above Seidel switching, then

$$S(G') = DS(G)D. \tag{1}$$

Indeed, if two vertices belong to the same part, then $d_i d_j = 1$, so the corresponding entry of the Seidel matrix is unchanged. If they belong to different parts, then $d_i d_j = -1$, so the corresponding Seidel entry changes sign, which is exactly the effect of reversing adjacency between the two parts.

Since $D^T = D$ and $D^2 = I$, the matrices $S(G)$ and $S(G')$ are orthogonally similar. Therefore,

$$\mathrm{Spec}(S(G')) = \mathrm{Spec}(S(G)).$$

Consequently, Seidel switching preserves the $p$-Seidel energy for every $p > 0$:

$$\mathcal{E}_p(S(G')) = \mathcal{E}_p(S(G)).$$

Two graphs are called *$S$-equivalent* if one can be obtained from the other by a Seidel switching. They are called *$SC$-equivalent* if one graph is $S$-equivalent to either the other graph or its complement [8].

In this article, we study extremal problems for the $p$-Seidel energy over graphs and regular graphs of fixed order. The paper is organized as follows.

In Section 2, for $p > 2$, we establish a universal lower bound for the $p$-Seidel energy of graphs of order $n$ and characterize its equality case in terms of symmetric conference Seidel matrices. Consequently, whenever a symmetric conference matrix of order $n$ exists, this bound determines the exact minimum $p$-Seidel energy and all graphs attaining it. For $0 < p < 2$, we obtain the corresponding upper bound and equality characterization. Hence, whenever a symmetric conference matrix of order $n$ exists, the exact maximum $p$-Seidel energy is determined. When no such matrix exists, the corresponding inequality is strict. In Section 3, for $p > 2$, $n \equiv 1 \pmod 4$, and $r = \frac{n-1}{2}$, we establish a lower bound for the $p$-Seidel energy among all $r$-regular graphs of order $n$ and characterize its equality case. Consequently, whenever a conference graph of order $n$ exists, this bound determines the minimum $p$-Seidel energy and all graphs attaining it. In particular, when $n$ is a prime power with $n \equiv 1 \pmod 4$, the Paley graph $P(n)$ is a conference graph and hence attains this minimum. In Section 4, we characterize the graph that achieves the maximum $p$-Seidel energy among all $r$-regular graphs with fixed order $n = 2r$, where $p \geq 3$. We also determine the maximum $p$-Seidel energy among $r$-regular graphs of order $2r + 2$ for $p \geq 3$. In this case, the maximum is attained uniquely by the disjoint union of two complete graphs of order $r + 1$. In Section 5, we state open problems concerning the $p$-Seidel energy for different values of $p$.

## 2 A lower bound for the $p$-Seidel energy, $p > 2$

Haemers studied the concept of Seidel energy of a graph, and he proposed the conjecture, which is later became known as *Haemers' Conjecture*, i.e., the complete graph has minimum Seidel energy among all graphs with $n$ vertices [8]. Akbari, Einollahzadeh, Karkhaneei, and Nematollahi proved the conjecture and showed that equality is characterized up to $SC$-equivalence [1]. Also in 2024, Einollahzadeh and Nematollahi gave a short proof of *Haemers' Conjecture* on the Seidel energy of graphs [7]. In particular, the complete graph is not the unique equality graph up to isomorphism.

**Theorem 2.1.** [1, Theorem 2] Let $G$ be a graph with $n$ vertices. Then

$$\mathcal{E}(S(G)) \geq \mathcal{E}(K_n),$$

and equality holds if and only if $G$ is $SC$-equivalent to $K_n$.

Also, in [1], the following strict lower bound of $p$-Seidel energy of graphs for any $p \in (0, 2)$ is established.

**Theorem 2.2.** [1, Theorem 1] Let $G$ be a graph with $n$ vertices. Then, for every real number $p \in (0, 2)$

$$\mathcal{E}_p(S(G)) > (n-1)^p + (n-2).$$

*Remark* 2.3. Since $S(G)$ is a real symmetric matrix, its squared Frobenius norm is

$$\|S(G)\|_F^2 = \operatorname{tr}(S(G)^\top S(G)) = \operatorname{tr}(S(G)^2).$$

The diagonal entries of $S(G)$ are zero and all its off-diagonal entries are equal to $\pm 1$. Therefore,

$$\|S(G)\|_F^2 = \sum_{i,j=1}^{n} |S(G)_{ij}|^2 = n(n-1).$$

On the other hand, if $\lambda_1, \ldots, \lambda_n$ are the Seidel eigenvalues of $G$, then

$$\operatorname{tr}(S(G)^2) = \sum_{i=1}^{n} \lambda_i^2.$$

Consequently,

$$\sum_{i=1}^{n} \lambda_i^2 = \|S(G)\|_F^2 = n(n-1). \tag{2}$$

Thus, the Euclidean norm of the Seidel spectrum is fixed for every graph of order $n$. In particular,

$$\mathcal{E}_2(S(G)) = \sum_{i=1}^{n} |\lambda_i|^2 = n(n-1),$$

so the 2-Seidel energy is independent of the structure of $G$. Therefore, the extremal $p$-Seidel energy problem is nontrivial only for $0 < p < 2$ or $p > 2$.

The discussion in Remark 2.3 leads to the following question:

**Question 2.4.** *What is the minimum $p$-Seidel energy among all graphs of order $n$, for $p > 2$, and which graphs attain it?*

To answer Question 2.4, we first recall the definition of a conference matrix.

**Definition 2.5.** [8, Page 658] A *conference matrix* is a square matrix $S$ of order $n$ with zero diagonal and $\pm 1$ off-diagonal entries such that

$$SS^\top = (n-1)I.$$

We say that a graph $G$ has a *symmetric conference Seidel matrix* if its Seidel matrix $S(G)$ is a symmetric conference matrix.

*Remark* 2.6. Conference matrices were introduced by Belevitch in connection with ideal telephone conference networks [3]. Throughout this article, whenever a symmetric conference Seidel matrix of order $n$ is considered, we assume that $n$ is an order for which such a matrix exists. The subsequent combinatorial formulation linked these matrices with conference graphs [14].

The following lemma is known as the classical power mean inequality (also called the generalized mean inequality).

**Lemma 2.7.** [5, Theorem 1] Let $a_1, a_2, \ldots, a_n$ be nonnegative real numbers and let $r > s > 0$. Then

$$\left(\frac{1}{n}\sum_{i=1}^{n} a_i^r\right)^{\frac{1}{r}} \geq \left(\frac{1}{n}\sum_{i=1}^{n} a_i^s\right)^{\frac{1}{s}},$$

with equality if and only if $a_1 = a_2 = \cdots = a_n$.

In the following theorem, we establish a universal lower bound for the $p$-Seidel energy and characterize its equality case. Consequently, for every order $n$ admitting a symmetric conference matrix, the theorem determines the minimum $p$-Seidel energy, which also answers Question 2.4 for such $n$.

**Theorem 2.8.** *Let $G$ be a graph on $n \geq 2$ vertices. Then, for every $p > 2$,*

$$\mathcal{E}_p(S(G)) \;\geq\; n\,(n-1)^{\frac{p}{2}},$$

*with equality if and only if $S(G)^2 = (n-1)I$, that is, if and only if $S(G)$ is a symmetric conference matrix.*

*Proof.* Let $\lambda_i$ be the Seidel eigenvalues of $G$, for $i = 1, 2, \ldots, n$.

Substituting $a_i = |\lambda_i|$ into Lemma 2.7 and using (2) we obtain

$$\mathcal{E}_p(S(G)) = \sum_{i=1}^{n} |\lambda_i|^p = \sum_{i=1}^{n} a_i^p \;\geq\; n\left(\frac{1}{n}\sum_{i=1}^{n} a_i^2\right)^{\frac{p}{2}} = n\left(\frac{1}{n}\sum_{i=1}^{n} \lambda_i^2\right)^{\frac{p}{2}} = n\,(n-1)^{\frac{p}{2}},$$

equality holds if and only if $|\lambda_1| = |\lambda_2| = \cdots = |\lambda_n|$.

Suppose that equality holds, and let $|\lambda_1| = \cdots = |\lambda_n| = a$.

By (2), $na^2 = \sum_{i=1}^{n} \lambda_i^2 = n(n-1)$, and hence $a^2 = n-1$. Therefore, $\lambda_i^2 = n-1$, $i = 1, \ldots, n$.

Since $S(G)$ is real symmetric, there exists an orthogonal matrix $Q$ such that

$$S(G) = Q\Lambda Q^\top,$$

where $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$.

Consequently,

$$S(G)^2 = Q\Lambda^2 Q^\top = Q((n-1)I)Q^\top = (n-1)I.$$

Since $S(G)$ is symmetric, has zero diagonal, and has $\pm 1$ off-diagonal entries, it follows that $S(G)$ is a symmetric conference matrix.

Conversely, suppose that $S(G)^2 = (n-1)I$. If $\lambda_i$ is an eigenvalue of $S(G)$, then $\lambda_i^2 = n-1$, and therefore

$$|\lambda_i| = \sqrt{n-1} \qquad \text{for all } i = 1, \ldots, n.$$

Hence

$$\mathcal{E}_p(S(G)) = \sum_{i=1}^{n} |\lambda_i|^p = n(n-1)^{\frac{p}{2}}.$$

Thus equality holds if and only if $S(G)$ is a symmetric conference matrix.

This completes the proof. □

*Remark* 2.9. The lower bound in Theorem 2.8 is attained only for those orders $n$ for which a symmetric conference matrix of order $n$ exists. Indeed, if no such matrix exists, then the equality characterization in Theorem 2.8 implies that

$$\mathcal{E}_p(S(G)) > n(n-1)^{\frac{p}{2}}$$

for every graph $G$ of order $n$.

There are well-known necessary arithmetic conditions for the existence of a symmetric conference matrix. In particular, for $n > 2$, if a symmetric conference matrix of order $n$ exists, then

$$n \equiv 2 \pmod 4$$

and

$$n - 1 = a^2 + b^2$$

for some integers $a$ and $b$; see, for example, [6, Lemma 10.4.1 and Theorem 10.4.2, p. 158]. Thus, if either $n \not\equiv 2 \pmod 4$ or $n-1$ is not representable as a sum of two integer squares, then no symmetric conference matrix of order $n$ exists, and hence the inequality in Theorem 2.8 is strict.

*Remark* 2.10. For $0 < p < 2$, let $\lambda_1, \ldots, \lambda_n$ be the Seidel eigenvalues of a graph $G$ of order $n$. Applying Lemma 2.7 to $a_i = |\lambda_i|$ and using (2), we obtain

$$\mathcal{E}_p(S(G)) = \sum_{i=1}^{n} |\lambda_i|^p = \sum_{i=1}^{n} a_i^p \;\le\; n \left( \frac{1}{n} \sum_{i=1}^{n} a_i^2 \right)^{\frac{p}{2}} = n \left( \frac{1}{n} \sum_{i=1}^{n} \lambda_i^2 \right)^{\frac{p}{2}} = n\,(n-1)^{\frac{p}{2}},$$

Equality holds if and only if $|\lambda_1| = \cdots = |\lambda_n| = \sqrt{n-1}$. Since $S(G)$ is real symmetric, this is equivalent to $S(G)^2 = (n-1)I$, that is, $S(G)$ is a symmetric conference matrix.

Consequently, if a symmetric conference matrix of order $n$ exists, then

$$\max_{|V(G)|=n} \mathcal{E}_p(S(G)) = n(n-1)^{\frac{p}{2}},$$

and equality holds precisely for those graphs whose Seidel matrices are symmetric conference matrices.

If no symmetric conference matrix of order $n$ exists, then

$$\mathcal{E}_p(S(G)) < n(n-1)^{\frac{p}{2}}$$

for every graph $G$ of order $n$. In this case, the above inequality does not determine the exact maximum.

## 3 A lower bound for the $p$-Seidel energy of $r$-regular graphs, $p > 2$

**Lemma 3.1.** *Let $G$ be a graph with $n$ vertices. Let $p > 2$ and $\lambda_1, \lambda_2, \lambda_3, \ldots, \lambda_n$ be the Seidel eigenvalues of $G$. Then*

$$\sum_{i=2}^{n} |\lambda_i|^p \;\geq\; (n-1)^{1-\frac{p}{2}}\big(n(n-1)-\lambda_1^2\big)^{\frac{p}{2}}$$

*with equality if and only if $|\lambda_2| = \cdots = |\lambda_n|$.*

*Proof.* Let $t = n-1$ and $z_i = |\lambda_{i+1}|$ for $i = 1, \ldots, t$. Then we have

$$\sum_{i=2}^{n} |\lambda_i|^p = \sum_{i=1}^{t} z_i^p \quad \text{and} \quad \sum_{i=2}^{n} \lambda_i^2 = \sum_{i=1}^{t} z_i^2.$$

For $p > 2$ and using Lemma 2.7 we have

$$\left(\frac{1}{t}\sum_{i=1}^{t} z_i^p\right)^{\frac{1}{p}} \;\geq\; \left(\frac{1}{t}\sum_{i=1}^{t} z_i^2\right)^{\frac{1}{2}},$$

equality holds if and only if all $z_i$ are equal for $i = 1, \ldots, t$.

$$\Longrightarrow \sum_{i=1}^{t} z_i^p \;\geq\; t^{\,1-\frac{p}{2}} \left(\sum_{i=1}^{t} z_i^2\right)^{\frac{p}{2}},$$

equality holds if and only if all $z_i$ are equal for $i = 1, \ldots, t$.

$$\Longrightarrow \sum_{i=2}^{n} |\lambda_i|^p \;\geq\; (n-1)^{1-\frac{p}{2}} \left(\sum_{i=2}^{n} \lambda_i^2\right)^{\frac{p}{2}},$$

equality holds if and only if all $z_i = |\lambda_{i+1}|$ are equal for $i = 1, \ldots, (n-1)$.

Now using (2) we have $\sum\limits_{i=2}^{n} \lambda_i^2 = n(n-1) - \lambda_1^2$, which gives

$$\sum_{i=2}^{n} |\lambda_i|^p \;\geq\; (n-1)^{1-\frac{p}{2}}\big(n(n-1)-\lambda_1^2\big)^{\frac{p}{2}},$$

equality holds if and only if all $|\lambda_{i+1}|$ are equal for $i = 1, \ldots, (n-1)$, i.e., when $|\lambda_2| = |\lambda_3| = \cdots = |\lambda_n|$.

This completes the proof. □

**Lemma 3.2.** [6, Page 221] Let $G$ be an $r$-regular graph on $n$ vertices with adjacency eigenvalues $\theta_1 = r, \theta_2, \ldots, \theta_n$, where $\theta_1 = r$ corresponds to the eigenvector $\mathbf{1}$. Then the Seidel eigenvalue corresponding to $\mathbf{1}$ is $n-1-2r$, and the remaining Seidel eigenvalues are $-1-2\theta_i$, $i = 2, \ldots, n$.

**Theorem 3.3.** *Let $G$ be an $r$-regular graph on $n$ vertices and $\lambda_1, \lambda_2, \ldots, \lambda_n$ be the Seidel eigenvalues of $G$ such that $\lambda_1$ is the eigenvalue corresponding to $\mathbf{1}$. Then for every $p > 2$,*

$$\mathcal{E}_p(S(G)) \;\geq\; |n-1-2r|^p + (n-1)^{1-\frac{p}{2}}\big(n(n-1)-(n-1-2r)^2\big)^{\frac{p}{2}},$$

*with equality if and only if $|\lambda_2| = \cdots = |\lambda_n|$.*

*Proof.* By Lemma 3.2, since $\lambda_1$ is the Seidel eigenvalue corresponding to $\mathbf{1}$, we have

$$\lambda_1 = n - 1 - 2r.$$

The result now follows directly from Lemma 3.1. □

**Definition 3.4.** [6, Page 115] A graph $G$ with $n$ vertices is called a *strongly regular graph* with parameters $(n, k, \lambda, \mu)$, written as $G = \mathrm{srg}(n, k, \lambda, \mu)$, where $G$ is $k$-regular, any two adjacent vertices have exactly $\lambda$ common neighbors, and any two non-adjacent vertices have exactly $\mu$ common neighbors.

**Definition 3.5.** [6, Page 115] A strongly regular graph with parameters

$$\left(n, \frac{n-1}{2}, \frac{n-5}{4}, \frac{n-1}{4}\right)$$

is called a *conference graph*. Equivalently, a conference graph $G$ of order $n$ satisfies

$$G = \mathrm{srg}\left(n, \frac{n-1}{2}, \frac{n-5}{4}, \frac{n-1}{4}\right).$$

*Remark* 3.6. The notion of a conference graph above should not be confused with a graph whose Seidel matrix is itself a symmetric conference matrix. If $G$ is a conference graph of order $n$, then

$$S(G)\mathbf{1} = 0$$

and

$$S(G)^2 = nI - J.$$

Thus $S(G)$ is singular and is not a conference matrix.

The two notions are nevertheless closely related. After a suitable switching, a symmetric conference matrix of order $n+1$ can be written as

$$C = \begin{pmatrix} 0 & \mathbf{1}^\top \\ \mathbf{1} & S(G) \end{pmatrix},$$

where $G$ is a conference graph of order $n$.

**Definition 3.7.** [6, Page 115] Let $n$ be a prime power with $n \equiv 1 \pmod 4$, and let $\mathbb{F}_n$ be the finite field of order $n$. The *Paley graph* $P(n)$ is the graph with vertex set $\mathbb{F}_n$ in which two distinct vertices $x$ and $y$ are adjacent if and only if $x - y$ is a nonzero square in $\mathbb{F}_n$.

*Remark* 3.8. [6, Proposition 9.1.1] The Paley graph $P(n)$ is a conference graph. The adjacency spectrum of a conference graph $G$ is

$$\mathrm{Spec}(A(G)) = \left\{ \frac{n-1}{2}, \underbrace{\frac{-1+\sqrt{n}}{2}, \ldots, \frac{-1+\sqrt{n}}{2}}_{\frac{n-1}{2} \text{ times}}, \underbrace{\frac{-1-\sqrt{n}}{2}, \ldots, \frac{-1-\sqrt{n}}{2}}_{\frac{n-1}{2} \text{ times}} \right\}.$$

In the following theorem, for $p > 2$, $n \equiv 1 \pmod 4$, and $r = \frac{n-1}{2}$, we establish a lower bound for the $p$-Seidel energy among all $r$-regular graphs of order $n$ and characterize the equality case. Whenever a conference graph of order $n$ exists, this bound gives the exact minimum. In particular, if $n$ is a prime power, the Paley graph $P(n)$ attains this minimum.

**Theorem 3.9.** *Let $n \geq 5$ with $n \equiv 1 \pmod 4$, and let $r = \frac{n-1}{2}$. Let $G$ be an $r$-regular graph on $n$ vertices. Then, for every $p > 2$,*

$$\mathcal{E}_p(S(G)) \geq (n-1)n^{\frac{p}{2}}.$$

*Equality holds if and only if $G$ is a conference graph. In particular, whenever $n$ is a prime power, the Paley graph $P(n)$ attains equality.*

*Proof.* Label the Seidel eigenvalues of $G$ as $\lambda_1, \lambda_2, \ldots, \lambda_n$ so that $\lambda_1$ is the eigenvalue corresponding to $\mathbf{1}$. Since $G$ is $r$-regular, Lemma 3.2 gives

$$\lambda_1 = n - 1 - 2r.$$

Since $r = \frac{n-1}{2}$, we have $\lambda_1 = 0$. Therefore, by Theorem 3.3,

$$\begin{aligned} \mathcal{E}_p(S(G)) &\geq (n-1)^{1-\frac{p}{2}} \big(n(n-1)\big)^{\frac{p}{2}} \\ &= (n-1)n^{\frac{p}{2}}. \end{aligned} \tag{3}$$

Equality holds if and only if $|\lambda_2| = |\lambda_3| = \cdots = |\lambda_n|$. Suppose first that equality holds. Then there exists $\alpha > 0$ such that $\lambda_i \in \{\alpha, -\alpha\}$, $i = 2, \ldots, n$.

Let $m_+$ and $m_-$ denote the multiplicities of $\alpha$ and $-\alpha$, respectively. Since $\operatorname{tr}(S(G)) = \sum_{i=1}^{n} \lambda_i = 0$ and $\lambda_1 = 0$, we obtain

$$\alpha(m_+ - m_-) = 0.$$

Hence

$$m_+ = m_- = \frac{n-1}{2}.$$

Furthermore, using (2),

$$n(n-1) = \sum_{i=1}^{n} \lambda_i^2 = (m_+ + m_-)\alpha^2 = (n-1)\alpha^2.$$

Thus $\alpha^2 = n$, and consequently

$$\operatorname{Spec}(S(G)) = \left\{ 0,\ \underbrace{-\sqrt{n}, \ldots, -\sqrt{n}}_{\frac{n-1}{2} \text{ times}},\ \underbrace{\sqrt{n}, \ldots, \sqrt{n}}_{\frac{n-1}{2} \text{ times}} \right\}.$$

We now show that this equality condition implies that $G$ is a conference graph.

Let $r = \frac{n-1}{2}$ be the trivial adjacency eigenvalue of $G$. By Lemma 3.2, the remaining adjacency eigenvalues are obtained from

$$\lambda = -1 - 2\theta.$$

Hence they are

$$\theta_+ = \frac{-1 + \sqrt{n}}{2}, \qquad \theta_- = \frac{-1 - \sqrt{n}}{2},$$

each with multiplicity $\frac{n-1}{2}$.

Let $A = A(G)$. On the subspace orthogonal to $\mathbf{1}$, the two nontrivial eigenvalues satisfy

$$x^2 + x - \frac{n-1}{4} = 0.$$

Let $M = A^2 + A - \frac{n-1}{4}I$. As shown above, $M$ vanishes on $\mathbf{1}^\perp$, while $M\mathbf{1} = \frac{n(n-1)}{4}\mathbf{1}$.

Since $\mathbb{R}^n = \operatorname{span}\{\mathbf{1}\} \oplus \mathbf{1}^\perp$ and the orthogonal projection onto $\operatorname{span}\{\mathbf{1}\}$ is $\frac{1}{n}J$, it follows that

$$M = \frac{n(n-1)}{4}\left(\frac{1}{n}J\right) = \frac{n-1}{4}J.$$

Hence

$$A^2 + A - \frac{n-1}{4}I = \frac{n-1}{4}J,$$

or equivalently,

$$A^2 = \frac{n-1}{4}I - A + \frac{n-1}{4}J. \tag{4}$$

For distinct vertices $i$ and $j$, the entry $(A^2)_{ij}$ is the number of their common neighbors. If $i$ and $j$ are adjacent, then $A_{ij} = 1$, and (4) gives

$$(A^2)_{ij} = -1 + \frac{n-1}{4} = \frac{n-5}{4}.$$

If $i$ and $j$ are nonadjacent, then $A_{ij} = 0$, and hence

$$(A^2)_{ij} = \frac{n-1}{4}.$$

Thus

$$G = \operatorname{srg}\left(n, \frac{n-1}{2}, \frac{n-5}{4}, \frac{n-1}{4}\right),$$

so $G$ is a conference graph.

Conversely, suppose that $G$ is a conference graph. Then

$$G = \operatorname{srg}\left(n, \frac{n-1}{2}, \frac{n-5}{4}, \frac{n-1}{4}\right).$$

The adjacency spectrum of $G$ is

$$\operatorname{Spec}(A(G)) = \left\{\frac{n-1}{2}, \underbrace{\frac{-1+\sqrt{n}}{2}, \ldots, \frac{-1+\sqrt{n}}{2}}_{\frac{n-1}{2}\text{ times}}, \underbrace{\frac{-1-\sqrt{n}}{2}, \ldots, \frac{-1-\sqrt{n}}{2}}_{\frac{n-1}{2}\text{ times}}\right\}.$$

By Lemma 3.2, its Seidel spectrum is therefore

$$\operatorname{Spec}(S(G)) = \left\{0, \underbrace{-\sqrt{n}, \ldots, -\sqrt{n}}_{\frac{n-1}{2}\text{ times}}, \underbrace{\sqrt{n}, \ldots, \sqrt{n}}_{\frac{n-1}{2}\text{ times}}\right\}.$$

Consequently,

$$\begin{aligned}\mathcal{E}_p(S(G)) &= \frac{n-1}{2}|\sqrt{n}|^p + \frac{n-1}{2}|-\sqrt{n}|^p \\ &= (n-1)n^{\frac{p}{2}}.\end{aligned}$$

Hence equality holds if and only if $G$ is a conference graph.

Finally, when $n$ is a prime power with $n \equiv 1 \pmod 4$, the Paley graph $P(n)$ is a conference graph, and therefore it attains equality. □

*Remark* 3.10. The lower bound in Theorem 3.9 is attained only for those orders $n$ for which a conference graph of order $n$ exists. Indeed, if no such conference graph exists, then the equality characterization in Theorem 3.9 implies that

$$\mathcal{E}_p(S(G)) > (n-1)n^{\frac{p}{2}}$$

for every $\frac{n-1}{2}$-regular graph $G$ of order $n$.

Since there are only finitely many graphs of order $n$, the minimum $p$-Seidel energy over all $\frac{n-1}{2}$-regular graphs of order $n$ nevertheless exists; however, Theorem 3.9 does not determine its exact value when no conference graph of order $n$ exists.

In particular, when $n$ is a prime power with $n \equiv 1 \pmod 4$, the Paley graph $P(n)$ is a conference graph. Hence, for such orders $n$, the lower bound is attained and

$$\min \mathcal{E}_p(S(G)) = (n-1)n^{\frac{p}{2}},$$

where the minimum is taken over all $\frac{n-1}{2}$-regular graphs $G$ of order $n$.

# 4 Maximum $p$-Seidel energy of $r$-regular graphs, $p \geq 3$

In mathematical optimization, the method of Lagrange multipliers is a strategy for finding the local extrema (i.e., maxima and minima) of a function subject to constraint equations (i.e., subject to the condition that one or more equations have to be satisfied exactly by the chosen values of the variables) [10]. The following lemma summarizes the method of Lagrange multipliers.

**Lemma 4.1.** [12, Theorem, Page 327] Let $F : \mathbb{R}^n \to \mathbb{R}$ and $g_1, \ldots, g_k : \mathbb{R}^n \to \mathbb{R}$ be continuously differentiable functions, and consider the constrained optimization problem

$$\text{maximize/minimize } F(x) \quad \text{subject to} \quad g_1(x) = 0, \ \ldots, \ g_k(x) = 0.$$

Assume that $x^* \in \mathbb{R}^n$ is a local extremum and that the gradients $\nabla g_1(x^*), \ldots, \nabla g_k(x^*)$ are linearly independent. Then there exist real numbers $\alpha_1, \ldots, \alpha_k$ such that

$$\nabla F(x^*) = \alpha_1 \nabla g_1(x^*) + \cdots + \alpha_k \nabla g_k(x^*), \tag{5}$$

together with the constraints

$$g_1(x^*) = \cdots = g_k(x^*) = 0.$$

Thus, any interior constrained extremum must satisfy the system (5).

**Lemma 4.2.** [12, Section 11.5, p. 333] Let $F : \mathbb{R}^n \to \mathbb{R}$ and $g_1, \ldots, g_k : \mathbb{R}^n \to \mathbb{R}$ be twice continuously differentiable functions. Suppose that $x^* \in \mathbb{R}^n$ is a local maximum of $F$ subject to $g_1(x) = 0, \ldots, g_k(x) = 0$, and that the gradients

$$\nabla g_1(x^*), \ldots, \nabla g_k(x^*)$$

are linearly independent. Let $\alpha_1, \ldots, \alpha_k$ be the Lagrange multipliers satisfying

$$\nabla F(x^*) = \alpha_1 \nabla g_1(x^*) + \cdots + \alpha_k \nabla g_k(x^*).$$

Then, for every $u \in \mathbb{R}^n$ satisfying $\nabla g_j(x^*)^\top u = 0$, $j = 1, \ldots, k$, we have

$$u^\top \left( \nabla^2 F(x^*) - \sum_{j=1}^{k} \alpha_j \nabla^2 g_j(x^*) \right) u \leq 0.$$

To determine the maximum $p$-Seidel energy among all $r$-regular graphs, we first solve the following constrained optimization problem. Thus, the optimization problem with constraints is stated in the following lemma, which is valid for $p \geq 3$.

**Lemma 4.3.** *Let $n \geq 6$ and $p \geq 3$. Consider the following problem*

$$F = F(t_1, \ldots, t_{n-1}) = \sum_{i=1}^{n-1} |t_i|^p$$

*subject to the constraints*

$$\sum_{i=1}^{n-1} t_i = 1, \qquad \sum_{i=1}^{n-1} t_i^2 = n(n-1) - 1, \qquad |t_i| \leq n-1 \ \forall i = 1, \ldots, n-1.$$

*Then the maximum of $F$ is attained by all vectors that are $\{n-1, \underbrace{-1, -1, \ldots, -1}_{(n-2) \text{ times}}\}$ up to permutation.*

*Proof.* Let us consider the feasible set to be

$$\mathcal{M} = \Big\{(t_1, \ldots, t_{n-1}) \in \mathbb{R}^{n-1} : \sum_{i=1}^{n-1} t_i = 1, \ \sum_{i=1}^{n-1} t_i^2 = n(n-1) - 1, \ |t_i| \leq n-1 \ \forall i\Big\}.$$

Then $\mathcal{M}$ is non-empty as $t^{(0)} = \{n-1, \underbrace{-1, -1, \ldots, -1}_{(n-2) \text{ times}}\} \in \mathcal{M}$. The set $\mathcal{M}$ is closed and bounded, hence compact. Since $F$ is continuous, the maximum of $F$ over $\mathcal{M}$ is attained at some

$$t^* = (t_1^*, \ldots, t_{n-1}^*) \in \mathcal{M}.$$

**Case (1):** Suppose that $t^*$ has a coordinate on the boundary, i.e., $|t_j^*| = n-1$ for some $j$. Without loss of generality, assume that $j = z$, $1 \leq z \leq (n-1)$.

**Subcase (1.1):** Let $t_z^* = n-1$.

We have

$$\sum_{i \neq z, \ i=1}^{n-1} t_i^* = 1 - t_z^* = 1 - (n-1) = 2 - n.$$

Thus the arithmetic mean of $t_1^*, t_2^*, \ldots, t_{z-1}^*, t_{z+1}^*, \ldots, t_{n-1}^*$ is

$$\frac{1}{n-2} \sum_{i \neq z, \ i=1}^{n-1} t_i^* = \frac{2-n}{n-2} = -1.$$

We have

$$\sum_{i \neq z, \ i=1}^{n-1} (t_i^*)^2 = \big(n(n-1) - 1\big) - (t_z^*)^2 = n(n-1) - 1 - (n-1)^2 = n - 2.$$

Thus the arithmetic mean of the squares is

$$\frac{1}{n-2} \sum_{i \neq z, \ i=1}^{n-1} (t_i^*)^2 = \frac{n-2}{n-2} = 1.$$

By the Cauchy-Schwarz inequality,

$$\left(\sum_{i\neq z} t_i^*\right)^2 \leq (n-2)\sum_{i\neq z}(t_i^*)^2.$$

Since

$$\left(\sum_{i\neq z} t_i^*\right)^2 = (n-2)^2 = (n-2)\sum_{i\neq z}(t_i^*)^2.$$

Hence equality holds in the Cauchy-Schwarz inequality. Thus all $t_i^*$, $i \neq z$, are equal. Since

$$\sum_{i\neq z} t_i^* = 2-n,$$

we obtain $t_i^* = -1$, $i \neq z$. Therefore,

$$t^* \text{ is } \left\{ n-1, \underbrace{-1,\ldots,-1}_{n-2 \text{ times}} \right\} \text{ up to permutation.}$$

**Subcase (1.2):** Let $t_z^* = -(n-1)$.

We have

$$\sum_{i\neq z,\ i=1}^{n-1} t_i^* = 1 - t_z^* = 1 + (n-1) = n.$$

Thus the arithmetic mean of $t_1^*, t_2^*, \ldots, t_{z-1}^*, t_{z+1}^*, \ldots, t_{n-1}^*$ is

$$\frac{1}{n-2}\sum_{i\neq z,\ i=1}^{n-1} t_i^* = \frac{n}{n-2}.$$

Again we have

$$\sum_{i\neq z,\ i=1}^{n-1} (t_i^*)^2 = (n(n-1)-1) - (t_z^*)^2 = n(n-1) - 1 - (n-1)^2 = n-2.$$

Thus, the arithmetic mean of the squares is

$$\frac{1}{n-2}\sum_{i\neq z,\ i=1}^{n-1} (t_i^*)^2 = \frac{n-2}{n-2} = 1.$$

By the Cauchy-Schwarz inequality,

$$\left(\sum_{i\neq z} t_i^*\right)^2 \leq (n-2)\sum_{i\neq z}(t_i^*)^2.$$

Hence

$$n^2 \leq (n-2)^2,$$

which is impossible for $n \geq 3$. Hence, this subcase is impossible for a feasible point (a point that satisfies all the given constraints).

Combining the two subcases, we conclude that

$$t^* \text{ is } \left\{ n-1, \underbrace{-1, \ldots, -1}_{n-2 \text{ times}} \right\} \text{ up to permutation.}$$

**Case (2):** Assume that $|t_i^*| < n-1$, $i = 1, \ldots, n-1$. Thus $t^*$ lies in the interior with respect to the constraints $|t_i| \leq n-1$, while still satisfying the two equality constraints. Therefore, locally, we only need to consider the two equality constraints.

Let us define

$$g_1(t) = g_1(t_1, \ldots, t_{n-1}) = \sum_{i=1}^{n-1} t_i - 1, \quad g_2(t) = g_2(t_1, \ldots, t_{n-1}) = \sum_{i=1}^{n-1} t_i^2 - \big(n(n-1) - 1\big).$$

The gradients of the constraints are

$$\nabla g_1(t) = \big(\underbrace{1, 1, \ldots, 1}_{(n-1) \text{ times}}\big), \qquad \nabla g_2(t) = \big(\underbrace{2t_1, 2t_2, \ldots, 2t_{n-1}}_{(n-1) \text{ times}}\big).$$

The vectors $\nabla g_1(t^*)$ and $\nabla g_2(t^*)$ are linearly independent. Indeed, if they were linearly dependent, then $t_1^* = \cdots = t_{n-1}^*$. The first constraint would then give $t_i^* = \frac{1}{n-1}$ for every $i$, which contradicts the second constraint. Therefore, the hypotheses of Lemma 4.1 are satisfied.

Therefore, by Lemma 4.1 at $t^*$ there exist $\alpha_1,\ \alpha_2 \in \mathbb{R}$ such that

$$\nabla F(t^*) = \alpha_1 \nabla g_1(t^*) + \alpha_2 \nabla g_2(t^*), \tag{6}$$

together with the conditions

$$g_1(t^*) = 0, \quad g_2(t^*) = 0.$$

Now computing $\nabla F$ in each coordinate $t_j$ we have

$$\frac{\partial}{\partial t_j} |t_j|^p = p\, |t_j|^{p-1} \operatorname{sgn}(t_j).$$

Thus

$$\Big(\nabla F(t)\Big)_j = p\, t_j |t_j|^{p-2}.$$

Computing $\nabla g_1,\ \nabla g_2$ in each coordinate $t_j$ we have

$$\frac{\partial g_1}{\partial t_j} = 1, \quad \frac{\partial g_2}{\partial t_j} = 2t_j.$$

Substituting into equality (6), from the $j$-th coordinate of that vector equation, we obtain

$$p\, |t_j^*|^{p-1} \operatorname{sgn}(t_j^*) = \alpha_1 \cdot 1 + \alpha_2 \cdot 2t_j^* = \alpha_1 + 2\alpha_2 t_j^*.$$

Finally,

$$p\, |t_j^*|^{p-1} \operatorname{sgn}(t_j^*) - \alpha_1 - 2\alpha_2\, t_j^* = 0, \quad j = 1, \ldots, n-1. \tag{7}$$

We now prove that an interior maximizer can have at most two distinct coordinate values.

Define $f(x) = |x|^p$ and $h(x) = f'(x) - \alpha_1 - 2\alpha_2 x = p\, x|x|^{p-2} - \alpha_1 - 2\alpha_2 x$.

By (7), every coordinate $t_i^*$ is a zero of $h$. Moreover,

$$h'(x) = p(p-1)|x|^{p-2} - 2\alpha_2.$$

If $\alpha_2 \leq 0$, then $h$ is strictly increasing on $\mathbb{R}$. Hence, $h$ has at most one zero, which would imply that all coordinates $t_i^*$ are equal. This is impossible because of the two constraints. Therefore, $\alpha_2 > 0$.

Set $\rho = \left(\frac{2\alpha_2}{p(p-1)}\right)^{\frac{1}{p-2}}$. Then $h'(x) > 0 \quad \text{for } |x| > \rho, \qquad h'(x) < 0 \quad \text{for } |x| < \rho.$

Thus $h$ is increasing on $(-\infty, -\rho)$, decreasing on $(-\rho, \rho)$, and increasing on $(\rho, \infty)$. Hence $h$ has at most three distinct real zeros.

Let $u = (u_1, \ldots, u_{n-1})$ satisfy

$$\sum_{i=1}^{n-1} u_i = 0 \qquad \text{and} \qquad \sum_{i=1}^{n-1} t_i^* u_i = 0.$$

Since $\nabla g_1(t^*) = (1, \ldots, 1)$, $\nabla g_2(t^*) = 2(t_1^*, \ldots, t_{n-1}^*)$, these conditions are equivalent to

$$\nabla g_1(t^*)^\top u = 0, \qquad \nabla g_2(t^*)^\top u = 0.$$

Moreover, $\nabla^2 g_1 = 0$, $\nabla^2 g_2 = 2I$, and

$$\nabla^2 F(t^*) = \operatorname{diag}\left(p(p-1)|t_1^*|^{p-2}, \ldots, p(p-1)|t_{n-1}^*|^{p-2}\right).$$

Therefore, by Lemma 4.2,

$$\sum_{i=1}^{n-1} \left(p(p-1)|t_i^*|^{p-2} - 2\alpha_2\right) u_i^2 \leq 0. \tag{8}$$

Suppose that a value $x$ occurs at least twice among the coordinates of $t^*$. Choose two coordinates equal to $x$, and let $u$ have entries 1 and $-1$ at these two coordinates, respectively, and 0 elsewhere. Then

$$\sum_{i=1}^{n-1} u_i = 0 \qquad \text{and} \qquad \sum_{i=1}^{n-1} t_i^* u_i = 0.$$

Hence, by (8),

$$2\left(p(p-1)|x|^{p-2} - 2\alpha_2\right) \leq 0.$$

Therefore,

$$h'(x) = p(p-1)|x|^{p-2} - 2\alpha_2 \leq 0,$$

and consequently

$$|x| \leq \rho.$$

Since $h$ is strictly decreasing on $(-\rho, \rho)$, at most one distinct zero of $h$ lying in $[-\rho, \rho]$ can occur more than once among the coordinates of $t^*$.

We show by contradiction that three distinct coordinate values, say $a < b < c$, of $t^*$ are impossible. Since every coordinate value is a zero of $h$, and $h$ is strictly increasing on $(-\infty, -\rho)$, strictly decreasing on $(-\rho, \rho)$, and strictly increasing on $(\rho, \infty)$, the existence of three distinct zeros implies

$$a < -\rho < b < \rho < c.$$

Therefore,

$$h'(a) > 0, \qquad h'(b) < 0, \qquad h'(c) > 0.$$

Since every coordinate value that occurs at least twice must satisfy $h'(x) \le 0$, neither $a$ nor $c$ can occur more than once. Consequently, $a$ and $c$ occur exactly once, while $b$ occurs $n-3$ times.

Let $d_a = h'(a)$, $d_b = h'(b)$, $d_c = h'(c)$. Since $a < b < c$ are three distinct zeros of $h$, and $h$ is strictly increasing on $(-\infty, -\rho)$, strictly decreasing on $(-\rho, \rho)$, and strictly increasing on $(\rho, \infty)$, we have

$$a < -\rho < b < \rho < c.$$

Therefore,

$$d_a > 0, \qquad d_b < 0, \qquad d_c > 0.$$

Moreover, since $h(a) = h(b) = h(c) = 0$, we have

$$f'(b) - f'(a) = 2\alpha_2(b-a)$$

and

$$f'(c) - f'(b) = 2\alpha_2(c-b).$$

Hence

$$\frac{f'(b) - f'(a)}{b-a} = 2\alpha_2 = \frac{f'(c) - f'(b)}{c-b}.$$

Since $f''(x) = p(p-1)|x|^{p-2}$ is convex for $p \ge 3$, for every $s \in [0,1]$ we have

$$f''((1-s)a + sb) \le (1-s)f''(a) + sf''(b).$$

Integrating with respect to $s$ over $[0,1]$, we obtain

$$\int_0^1 f''((1-s)a + sb)\, ds \le \frac{f''(a) + f''(b)}{2}.$$

Using the substitution $x = (1-s)a + sb$, we get

$$\int_0^1 f''((1-s)a + sb)\, ds = \frac{1}{b-a}\int_a^b f''(x)\, dx.$$

Therefore,

$$\frac{1}{b-a}\int_a^b f''(x)\, dx \le \frac{f''(a) + f''(b)}{2}.$$

By the fundamental theorem of calculus,

$$\frac{1}{b-a}\int_a^b f''(x)\, dx = \frac{f'(b) - f'(a)}{b-a} = 2\alpha_2.$$

Hence

$$2\alpha_2 \le \frac{f''(a) + f''(b)}{2}.$$

Similarly,

$$2\alpha_2 \le \frac{f''(b) + f''(c)}{2}.$$

Since $d_a = f''(a) - 2\alpha_2$, $d_b = f''(b) - 2\alpha_2$, $d_c = f''(c) - 2\alpha_2$. Therefore, $d_a + d_b \ge 0$, $d_b + d_c \ge 0$, and consequently, $d_a \ge -d_b$, $d_c \ge -d_b$.

Let $m = n - 3 \ge 3$. Since the multiplicities of $a, b, c$ are $1, m, 1$, respectively, we choose

$$u = \left( c-b, \underbrace{-\frac{c-a}{m}, \ldots, -\frac{c-a}{m}}_{m \text{ times}}, b-a \right).$$

Then

$$\sum_{i=1}^{n-1} u_i = (c-b) - (c-a) + (b-a) = 0$$

and

$$\sum_{i=1}^{n-1} t_i^* u_i = a(c-b) - b(c-a) + c(b-a) = 0.$$

Hence $u$ is an admissible vector in (8). Therefore,

$$d_a(c-b)^2 + \frac{d_b}{m}(c-a)^2 + d_c(b-a)^2 \le 0.$$

On the other hand, using $d_a \ge -d_b$, $d_c \ge -d_b$, we obtain

$$d_a(c-b)^2 + \frac{d_b}{m}(c-a)^2 + d_c(b-a)^2 \ge -d_b\big((c-b)^2 + (b-a)^2\big) + \frac{d_b}{m}(c-a)^2.$$

Since $d_b < 0$ and $m \ge 3$,

$$\frac{d_b}{m} > \frac{d_b}{2}.$$

Hence

$$\begin{aligned} d_a(c-b)^2 + \frac{d_b}{m}(c-a)^2 + d_c(b-a)^2 &> -d_b\big((c-b)^2 + (b-a)^2\big) + \frac{d_b}{2}(c-a)^2 \\ &= -\frac{d_b}{2}\big((c-b) - (b-a)\big)^2 \ge 0, \end{aligned}$$

which contradicts (8).

Therefore, an interior maximizer cannot have three distinct coordinate values. Hence, $t^*$ has at most two distinct coordinate values.

Let $a$ and $b$ be the two distinct coordinate values of $t^*$. We claim that one of the two values $a$ and $b$ must occur exactly once. Suppose, to the contrary, that both $a$ and $b$ occur at least twice.

Choose two coordinates equal to $a$. Let $u$ have entries 1 and $-1$ at these two coordinates, respectively, and 0 elsewhere. Then

$$\sum_{i=1}^{n-1} u_i = 0$$

and

$$\sum_{i=1}^{n-1} t_i^* u_i = a - a = 0.$$

Hence $u$ is admissible in (8). Therefore, $2h'(a) \le 0$, and thus $h'(a) \le 0$. Applying the same argument to two coordinates equal to $b$, we obtain $h'(b) \le 0$.

By (7), every coordinate value of $t^*$ is a zero of $h$. Hence $h(a) = h(b) = 0$. Also, $h'(x) \le 0$ only for $|x| \le \rho$, and $h$ is strictly decreasing on $[-\rho, \rho]$. Therefore, $h$ can have at most one zero in this interval. But $a \ne b$, and both $a$ and $b$ are zeros of $h$ satisfying $h'(a) \le 0$ and $h'(b) \le 0$, which is impossible.

Therefore, one of the two values $a$ and $b$ must occur exactly once. Up to reordering, we may write

$$t_1^* = a, \qquad t_2^* = \cdots = t_{n-1}^* = b.$$

The constraints give

$$a + (n-2)b = 1, \tag{9}$$

$$a^2 + (n-2)b^2 = n(n-1) - 1. \tag{10}$$

From (9),

$$a = 1 - (n-2)b.$$

Substituting this into (10), we obtain

$$(1-(n-2)b)^2 + (n-2)b^2 = n(n-1) - 1.$$

Equivalently,

$$(n-2)(b+1)\big((n-1)b - (n+1)\big) = 0.$$

Therefore, $b = -1$ or $b = \frac{n+1}{n-1}$.

If $b = -1$, then $a = 1 - (n-2)(-1) = n-1$. Thus

$$t^* = \left\{ n-1, \underbrace{-1, \ldots, -1}_{n-2 \text{ times}} \right\},$$

which is precisely the boundary solution obtained in Case (1).

Now suppose that $b = \frac{n+1}{n-1}$. Then

$$a = 1 - (n-2)b = -(n-1) + \frac{2}{n-1}.$$

Let $M = n-1$, $A = M - \frac{2}{M}$, $B = 1 + \frac{2}{M}$. Then $|a| = A$, $b = B$, and, since $n \geq 6$,

$$M > A > B > 1.$$

Since both the vector $(M, -1, \ldots, -1)$ and $(a, b, \ldots, b)$ satisfy the same sum-of-squares constraint, we have

$$M^2 - A^2 = (M-1)(B^2 - 1). \tag{11}$$

Let $\varphi(x) = x^{\frac{p}{2}}$. Since $p \geq 3$, the derivative $\varphi'(x) = \frac{p}{2} x^{\frac{p}{2}-1}$ is strictly increasing on $(0, \infty)$.

By the Mean Value Theorem, there exist $\xi_1 \in (A^2, M^2)$ and $\xi_2 \in (1, B^2)$ such that

$$\frac{M^p - A^p}{M^2 - A^2} = \varphi'(\xi_1) \quad \text{and} \quad \frac{B^p - 1}{B^2 - 1} = \varphi'(\xi_2).$$

Since $\xi_1 > A^2 > B^2 > \xi_2$, we have

$$\varphi'(\xi_1) > \varphi'(\xi_2).$$

Therefore,

$$\frac{M^p - A^p}{M^2 - A^2} > \frac{B^p - 1}{B^2 - 1}.$$

Using (11), we obtain

$$M^p - A^p > (M-1)(B^p - 1).$$

Hence

$$A^p + (M-1)B^p < M^p + (M-1).$$

Therefore,

$$|a|^p + (n-2)|b|^p < (n-1)^p + (n-2).$$

Thus this second solution cannot be a global maximizer. Consequently, the only global maximizers are

$$t^* \text{ is } \left\{ n-1, \underbrace{-1, \ldots, -1}_{n-2 \text{ times}} \right\} \text{ up to permutation.}$$

This completes the proof. □

In the following result, Berman, Shaked-Monderer, Singh, and Zhang determine the Seidel spectrum of the complete bipartite graph $K_{a,b}$.

**Lemma 4.4.** [4, Lemma 2.1(c)] Let $K_{a,b}$ be the complete bipartite graph with $n = a + b$ vertices. Then the Seidel spectrum of $K_{a,b}$ is

$$\left\{ n-1, \underbrace{-1, \ldots, -1}_{n-1 \text{ times}} \right\}.$$

*Remark* 4.5. Lemma 4.4 is consistent with the equality statement in Theorem 2.1. Indeed, for every $a, b \geq 1$ with $a + b = n$,

$$\mathrm{Spec}(S(K_{a,b})) = \left\{ n-1, \underbrace{-1, \ldots, -1}_{n-1 \text{ times}} \right\}$$

and therefore

$$\mathcal{E}(S(K_{a,b})) = 2(n-1) = \mathcal{E}(S(K_n)).$$

Although $K_{a,b}$ is generally not isomorphic to $K_n$, it is $SC$-equivalent to $K_n$. Thus, the equality in the minimum Seidel-energy bound is an $SC$-equivalence characterization, rather than an isomorphism characterization.

**Lemma 4.6.** [11, Theorem 2.5.6] Every real symmetric matrix $A$ admits the spectral decomposition

$$A \;=\; \sum_{i=1}^{t} \lambda_i \, u_i u_i^{\top},$$

where $\lambda_1, \ldots, \lambda_t$ are the nonzero eigenvalues of $A$, and $u_1, \ldots, u_t$ are the corresponding orthonormal eigenvectors.

We now characterize the graphs whose Seidel spectrum is equal to Seidel spectrum of $K_{a,b}$, see Lemma 4.4, up to isomorphism.

**Lemma 4.7.** *Let $G$ be a graph on $n \geq 2$ vertices. Then the following statements are equivalent:*

1. $\mathrm{Spec}(S(G)) = \left\{ n-1, \underbrace{-1, -1, \ldots, -1}_{n-1 \text{ times}} \right\}$.
2. *There exists an integer $s$ with $1 \leq s \leq n-1$, such that $G \cong K_{s,n-s}$.*

*Proof.* (1) $\Rightarrow$ (2) : Suppose that

$$\mathrm{Spec}(S(G)) = \left\{ n-1, \underbrace{-1, -1, \ldots, -1}_{n-1 \text{ times}} \right\}.$$

Since $S(G)$ is real symmetric, by Lemma 4.6 it admits an orthonormal spectral decomposition. Let $u_1$ be a unit eigenvector corresponding to the eigenvalue $n-1$, and let $u_2, \ldots, u_n$ be orthonormal eigenvectors corresponding to the eigenvalue $-1$. Then

$$S(G) = (n-1)u_1u_1^\top - \sum_{i=2}^{n} u_iu_i^\top.$$

Since $\sum_{i=1}^{n} u_iu_i^\top = I$, we obtain

$$\begin{aligned} S(G) &= (n-1)u_1u_1^\top - \left(I - u_1u_1^\top\right) \\ &= -I + nu_1u_1^\top. \end{aligned}$$

Put $v = u_1$. Hence

$$S(G) = -I + nvv^\top.$$

Since every diagonal entry of a Seidel matrix is zero,

$$0 = S(G)_{ii} = -1 + nv_i^2,$$

and therefore

$$v_i^2 = \frac{1}{n} \qquad \text{for every } i = 1, \ldots, n.$$

Thus

$$v_i = \pm\frac{1}{\sqrt{n}}.$$

Define $z_i = \sqrt{n}\, v_i \in \{-1, 1\}$. For $i \neq j$,

$$S(G)_{ij} = nv_iv_j = z_iz_j.$$

Let $V_+ = \{\, v_i : z_i = 1 \,\}$, $V_- = \{\, v_i : z_i = -1 \,\}$. If two vertices $v_i, v_j$ belong to the same set, then $z_iz_j = 1$, and consequently

$$S(G)_{ij} = 1.$$

By the definition of the Seidel matrix, $v_i$ and $v_j$ are therefore non-adjacent. On the other hand, if $v_i \in V_+$ and $v_j \in V_-$, then $z_iz_j = -1$, and hence

$$S(G)_{ij} = -1.$$

Thus $v_i$ and $v_j$ are adjacent.

Consequently, there are no edges inside $V_+$ or $V_-$, while every vertex of $V_+$ is adjacent to every vertex of $V_-$. Therefore, $G$ is a complete bipartite graph.

Since $G$ has at least one edge, both $V_+$ and $V_-$ are nonempty. Let $|V_+| = s$, $|V_-| = n-s$ where $1 \leq s \leq n-1$. Then we obtain

$$G \cong K_{s,n-s}.$$

(2) $\Rightarrow$ (1) : Suppose that

$$G \cong K_{s,n-s}$$

for some $1 \leq s \leq n-1$. By Lemma 4.4, the Seidel spectrum of every complete bipartite graph on $n$ vertices is

$$\mathrm{Spec}(S(G)) = \left\{ n-1, \underbrace{-1, -1, \ldots, -1}_{n-1 \text{ times}} \right\}.$$

This completes the proof. □

**Corollary 4.8.** *Let $G$ be an $r$-regular graph on $n = 2r$ vertices. Then the following statements are equivalent:*

1. $\mathrm{Spec}(S(G)) = \{n-1, \underbrace{-1, -1, \ldots, -1}_{(n-1) \text{ times}}\}$.
2. $G \cong K_{r,r}$.

*Proof.* (1) $\Rightarrow$ (2) : Suppose that

$$\mathrm{Spec}(S(G)) = \left\{ n-1, \underbrace{-1, -1, \ldots, -1}_{(n-1) \text{ times}} \right\}.$$

By Lemma 4.7, there exists an integer $s$, with $1 \leq s \leq n-1$, such that

$$G \cong K_{s,n-s}.$$

In the complete bipartite graph $K_{s,n-s}$, every vertex in the part of size $s$ has degree $n-s$, while every vertex in the part of size $n-s$ has degree $s$.

Since $G$ is $r$-regular, all vertices of $G$ have the same degree $r$. Therefore, $s = r$, $n - s = r$. In particular, $s = n - s$. Hence $n = 2s$. Since $n = 2r$, it follows that $s = r$. Therefore,

$$G \cong K_{s,n-s} = K_{r,r}.$$

(2) $\Rightarrow$ (1) : Conversely, suppose that

$$G \cong K_{r,r}.$$

Since $n = 2r$, the graph $K_{r,r}$ is a complete bipartite graph on $n$ vertices. By Lemma 4.4, the Seidel spectrum of every complete bipartite graph $K_{s,n-s}$ is

$$\left\{ n-1, \underbrace{-1, -1, \ldots, -1}_{(n-1) \text{ times}} \right\}.$$

Therefore,

$$\mathrm{Spec}(S(G)) = \left\{ n-1, \underbrace{-1, -1, \ldots, -1}_{(n-1) \text{ times}} \right\}.$$

Hence (1) and (2) are equivalent. □

In the following theorem, we prove that among all $r$-regular graphs, the balanced complete bipartite graph (up to isomorphism) achieves the maximum $p$-Seidel energy with fixed order $n = 2r$, where $p \geq 3$.

**Theorem 4.9.** *Let $G$ be an $r$-regular graph on $n = 2r$ vertices. Then for every $p \geq 3$,*

$$\mathcal{E}_p(S(G)) \leq (n-1)^p + (n-1),$$

*with equality if and only if $G \cong K_{r,r}$.*

*Proof.* We first observe that every $r$-regular graph of order $n = 2r$ is connected. Indeed, every component of an $r$-regular simple graph contains at least $r+1$ vertices. If $G$ were disconnected, then it would have at least two components, and hence $n \geq 2(r+1) = 2r+2$, which contradicts $n = 2r$.

We first consider the cases $r = 1$ and $r = 2$. If $r = 1$, then $n = 2$, and therefore

$$G \cong K_{1,1}.$$

If $r = 2$, then $n = 4$. Since every component of a 2-regular simple graph is a cycle of length at least 3, a 2-regular graph on four vertices must be

$$G \cong C_4 \cong K_{2,2}.$$

Thus the result follows in both cases.

Hence, assume that $r \geq 3$.

Since $n = 2r$, we have $n \geq 6$, so Lemma 4.3 applies. Label the Seidel eigenvalues of $G$ as $\lambda_1, \lambda_2, \ldots, \lambda_n$ so that $\lambda_1$ is the eigenvalue corresponding to $\mathbf{1}$. Since $G$ is $r$-regular and $n = 2r$, Lemma 3.2 gives

$$\lambda_1 = n - 1 - 2r = -1.$$

Set $t_i = \lambda_{i+1}$, $i = 1, \ldots, n-1$. Since $\operatorname{tr}(S(G)) = 0$, we obtain

$$\sum_{i=1}^{n-1} t_i = -\lambda_1 = 1.$$

Moreover, using $\sum_{i=1}^{n} \lambda_i^2 = n(n-1)$, we have

$$\sum_{i=1}^{n-1} t_i^2 = n(n-1) - 1.$$

Also, every row of $S(G)$ has absolute row sum $n-1$. Hence $\|S(G)\|_\infty = \max_{1 \leq i \leq n} \sum_{j=1}^{n} |S(G)_{ij}| = n-1$. Since every eigenvalue $\lambda$ of a matrix satisfies $|\lambda| \leq \|S(G)\|_\infty$, every Seidel eigenvalue satisfies

$$|\lambda_i| \leq n-1, \qquad i = 1, \ldots, n.$$

Thus,

$$|t_i| \leq n-1, \qquad i = 1, \ldots, n-1.$$

Therefore, every $r$-regular graph $G$ of order $n = 2r$ determines a feasible point $(t_1, \ldots, t_{n-1})$ of the optimization problem in Lemma 4.3. Consequently,

$$\begin{aligned}
\mathcal{E}_p(S(G)) &= |\lambda_1|^p + \sum_{i=1}^{n-1} |t_i|^p \\
&= 1 + \sum_{i=1}^{n-1} |t_i|^p \\
&\leq 1 + \max_{\mathcal{M}} \sum_{i=1}^{n-1} |t_i|^p,
\end{aligned}$$

where $\mathcal{M}$ is the feasible set in Lemma 4.3.

By Lemma 4.3,

$$\max_{\mathcal{M}} \sum_{i=1}^{n-1} |t_i|^p = (n-1)^p + (n-2).$$

Hence

$$\mathcal{E}_p(S(G)) \leq (n-1)^p + (n-1).$$

Now suppose that equality holds, that is,

$$\mathcal{E}_p(S(G)) = (n-1)^p + (n-1).$$

Then the associated feasible vector $(t_1, \ldots, t_{n-1})$ must attain the maximum in Lemma 4.3. Hence, up to ordering,

$$(t_1, \ldots, t_{n-1}) = \left( n-1, \underbrace{-1, \ldots, -1}_{n-2 \text{ times}} \right).$$

Since $\lambda_1 = -1$, we obtain

$$\mathrm{Spec}(S(G)) = \left\{ n-1, \underbrace{-1, -1, \ldots, -1}_{(n-1) \text{ times}} \right\}.$$

Therefore, by Corollary 4.8,

$$G \cong K_{r,r}.$$

Conversely, suppose that $G \cong K_{r,r}$.

Since $n = 2r$, the graph $K_{r,r}$ is an $r$-regular graph of order $n$. By Corollary 4.8,

$$\mathrm{Spec}(S(G)) = \left\{ n-1, \underbrace{-1, -1, \ldots, -1}_{(n-1) \text{ times}} \right\}.$$

Hence

$$\mathcal{E}_p(S(G)) = |n-1|^p + (n-1)| - 1|^p = (n-1)^p + (n-1).$$

Thus equality holds. □

*Remark* 4.10. For $n = 2r$, we have $\mathcal{E}_p(S(K_n)) = \mathcal{E}_p(S(K_{r,r})) = (n-1)^p + (n-1)$.

Indeed,

$$\mathrm{Spec}(S(K_n)) = \left\{ -(n-1), \underbrace{1, 1, \ldots, 1}_{(n-1) \text{ times}} \right\},$$

whereas

$$\mathrm{Spec}(S(K_{r,r})) = \left\{ n-1, \underbrace{-1, -1, \ldots, -1}_{(n-1) \text{ times}} \right\}.$$

Thus, the Seidel eigenvalues of $K_n$ and $K_{r,r}$ differ only by sign, and hence their absolute values are the same. Therefore, they have the same $p$-Seidel energy.

This does not contradict Theorem 4.9, since the theorem considers only $r$-regular graphs of order $n = 2r$. The graph $K_n$ is $(n-1)$-regular and, for $n > 2$, does not belong to this class. Hence, within the class considered in Theorem 4.9, equality is attained if and only if $G \cong K_{r,r}$.

Before proving the next result, we first note a useful relation between the Seidel matrices of a graph and its complement. Since $A(\overline{G}) = J - I - A(G)$, we have

$$\begin{aligned} S(\overline{G}) &= J - I - 2A(\overline{G}) \\ &= J - I - 2\big(J - I - A(G)\big) \\ &= -J + I + 2A(G) \\ &= -S(G). \end{aligned}$$

Therefore, if $\lambda_1, \lambda_2, \ldots, \lambda_n$ are the Seidel eigenvalues of $G$, then $-\lambda_1, -\lambda_2, \ldots, -\lambda_n$ are the Seidel eigenvalues of $\overline{G}$. Consequently, for every $p > 0$,

$$\mathcal{E}_p(S(\overline{G})) = \sum_{i=1}^{n} |-\lambda_i|^p = \sum_{i=1}^{n} |\lambda_i|^p = \mathcal{E}_p(S(G)).$$

**Theorem 4.11.** *Let $r \geq 1$, $n = 2r+2$, and $p \geq 3$. Among all $r$-regular graphs $G$ of order $n$,*

$$\mathcal{E}_p(S(G)) \leq (n-1)^p + (n-1).$$

*Moreover, equality holds if and only if $G \cong K_{r+1} \cup K_{r+1}$.*

*Proof.* Let $G$ be an $r$-regular graph of order $n = 2r + 2$, and put $H = \overline{G}$. Since every vertex of $G$ has degree $r$, every vertex of $H$ has degree $n-1-r = (2r+2)-1-r = r+1$. Thus, $H$ is an $(r+1)$-regular graph of order $n = 2(r+1)$.

Put $s = r+1$. Then $H$ is a $s$-regular graph of order $n = 2s$. Therefore, by Theorem 4.9,

$$\mathcal{E}_p(S(H)) \leq (n-1)^p + (n-1),$$

with equality if and only if $H \cong K_{s,s}$.

Since $S(H) = S(\overline{G}) = -S(G)$, we have

$$\mathcal{E}_p(S(H)) = \mathcal{E}_p(S(G)).$$

Hence

$$\mathcal{E}_p(S(G)) \leq (n-1)^p + (n-1).$$

Now suppose that equality holds. Then equality also holds for $H$, and hence, by Theorem 4.9,

$$H \cong K_{s,s} = K_{r+1,r+1}.$$

Since $H = \overline{G}$, taking complements gives

$$G \cong \overline{K_{r+1,r+1}} = K_{r+1} \cup K_{r+1}.$$

Conversely, suppose that $G \cong K_{r+1} \cup K_{r+1}$. Then

$$H = \overline{G} \cong K_{r+1,r+1} = K_{s,s}.$$

By the equality case of Theorem 4.9,

$$\mathcal{E}_p(S(H)) = (n-1)^p + (n-1).$$

Since $\mathcal{E}_p(S(G)) = \mathcal{E}_p(S(H))$, we obtain

$$\mathcal{E}_p(S(G)) = (n-1)^p + (n-1).$$

Therefore, equality holds if and only if

$$G \cong K_{r+1} \cup K_{r+1}.$$

□

## 5 Some open problems

We conclude with the following open questions.

**Question 5.1.** *Let $p > 2$. Characterize the graphs that maximize the $p$-Seidel energy among all graphs with fixed order $n$.*

**Question 5.2.** *Let $0 < p < 2$. Characterize the graphs that minimize the $p$-Seidel energy among all graphs with fixed order $n$.*

**Question 5.3.** *Let $0 < p < 2$. Characterize the $r$-regular graphs that maximize (or minimize) the $p$-Seidel energy among all $r$-regular graphs with fixed order $n$.*

**Question 5.4.** *Let $p \geq 3$. For the parameter pairs $(n, r)$ not covered by Theorems 4.9 and 4.11, characterize the $r$-regular graphs that maximize the $p$-Seidel energy among all $r$-regular graphs of order $n$.*

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## References

[1] S. Akbari, M. Einollahzadeh, M. M. Karkhaneei, and M. A. Nematollahi, *Proof of a conjecture on the Seidel energy of graphs*, European Journal of Combinatorics, (86) (2020), 103078.

[2] S Akbari, H Kumar, B Mohar, and S Pragada, *A Linear Lower Bound for the Square Energy of Graphs*, The Electronic Journal of Combinatorics, 32 (3), (2025), P3.53. doi.org/10.37236/13467

[3] V. Belevitch, *Theorem of $2n$-terminal networks with application to conference telephony*, Electrical Communication, (26) (1950), 231-244.

[4] A. Berman, N. Shaked-Monderer, R. Singh, and X. D. Zhang, Complete multipartite graphs that are determined, up to switching, by their Seidel spectrum, Linear Algebra and Its Applications, (564) (2019), 58-71.

[5] P. S. Bullen, Handbook of means and their inequalities, Springer Science and Business Media, (2013) Apr 17.

[6] A. E. Brouwer, and W. H. Haemers, Spectra of graphs, Springer Science and Business Media, (2011) Dec 17.

[7] M. Einollahzadeh and M. A. Nematollahi, *A short proof of Haemers' conjecture on the Seidel energy of graphs*, Linear Algebra and Its Applications, (695) (2024), 75-88.

[8] W. H. Haemers, *Seidel switching and graph energy*, MATCH Commun. Math. Comput. Chem., (68) (2012), 653–659.

[9] W. H. Haemers and Q. Xiang, *Strongly regular graphs with parameters* $(4m^4, 2m^4 + m^2, m^4 + m^2, m^4 + m^2)$ *exist for all* $m>1$, European Journal of Combinatorics, (31) (6) (2010), 1553-1559.

[10] L. D. Hoffmann, G. L. Bradley, K. H. Rosen, *Calculus for business, economics, and the social and life sciences*, New York, USA: McGraw-Hill, 1989 Jan.

[11] R. A. Horn and C. R. Johnson, *Matrix Analysis*, Cambridge University Press, 2nd ed., 2013.

[12] D. G. Luenberger and Y. Ye, *Linear and Nonlinear Programming*, 3rd ed., Springer, 2008.

[13] J. H. Vant Lint, J. J. Seidel, *Equilateral point sets in elliptic geometry*, Proceedings of the Koninklijke Nederlandse Akademie van Wetenschappen: Series A: Mathematical Sciences, (3) (69) (1966), 335-348.

[14] J. H. van Lint, R. M. Wilson, *A Course in Combinatorics*, Cambridge University Press, Cambridge, 2001.